

REVIEW #1 (Category A)

Summary: Development of a method for spatial standardization of football metrics (heatmaps) based on the Adaptive Multi-Interval Scale (AMIS). An algorithm for the nonlinear transformation of game event density into a 0–100 interval scale is proposed, with the benchmark fixed on data from the 2012/13 Russian Championship (32 team performances, 32,000 observations). A global normalized heatmap is presented as a measurement instrument.

Relevance: High. The problem of spatial data incomparability in sports analytics represents a fundamental methodological barrier hindering the integration of heterogeneous parameters.

Scientific Novelty (28/30): The work demonstrates characteristics of **P-Novelty**:

- a) **Logical integrity:** a consistent transition from critique of existing methods (Z-score, Min-Max) to the construction of a new measurement axiomatics.
- b) **Explanation of limitations:** the "statistical relativism" of traditional heatmaps is articulated.
- c) **Research program potential:** the possibility of extending the method to medicine, urban studies, and flaw detection is stated.

Reduction to 28 due to the absence of formal proof of convergence of the AMIS algorithm to the true distribution (empirical validation was conducted, but without mathematical rigor).

Methodological Rigor (24/25): For P-Novelty, adequacy is assessed:

- a) **Consistency:** the algorithm is described step-by-step with formulas and control points.
- b) **Data scale:** 32,000 observations constitute a representative base for calibration.
- c) **Transparency:** open repositories (GitHub, Dataverse), availability of code and data.

Practical Value (19/20): High potential for integration into sports analytics systems (StatsBomb, Catapult). The ability to conduct retrospective analysis of historical matches (since 1958) is a unique advantage.

Visualization Quality (14/15): 7 illustrations (Figs. 1–7) with correct use of color scales and smoothing. Numerical grid values are present (Fig. 7), enhancing evidentiary value.

Ethical Aspects (10/10): Full compliance with FAIR principles (Findable, Accessible, Interoperable, Reusable). Data in Dataverse, code open, no conflict of interest declared.

Final Score: $28 + 24 + 19 + 14 + 10 = 95/100$

Recommendation: Accept (highest standard, 90–100). Journal level — Q1.

Document Type: Fundamental work (Category A).

REVIEW 1 (in descending order of scientific value)

Article 4: "METHODOLOGY FOR STANDARDIZATION OF FOOTBALL METRICS HEATMAPS USING THE ADAPTIVE MULTI-INTERVAL SCALE (AMIS) part 1"

Summary: A fundamentally new method for spatial standardization of game activity in football is proposed — the Adaptive Multi-Interval Scale (AMIS), which allows transforming raw data into a universal 100-point interval scale with a fixed reference point (50 points = benchmark level of the 2012/13 Russian Championship). In the first part of the work, a global benchmark of activity distribution on a 40×25 cell grid is constructed, validated on data from 32 team performances.

Relevance: High. The problem of spatial data incomparability in sports analytics (different eras, leagues, measurement systems) is a fundamental methodological barrier. The proposed solution has the potential to become an industry standard.

Scientific Novelty (30%): P-Novelty (28/30). The work demonstrates:

- a) **Logical integrity** — the AMIS algorithm is sequentially described from formal definition to software implementation (Python, Excel plugin); a unified transformation function is constructed on 32,000 observations.
- b) **Explanation of limitations of traditional approaches** — the incorrectness of Z-score (normality requirement) and Min-Max (sensitivity to outliers) for nonlinear activity distributions is shown in detail.
- c) **Potential for a new research program** — the method extends to 2D space, opening the possibility of integrating heterogeneous parameters (speed, technical-tactical) into a unified metric system.

Justification: The concept of a metrological standard for the football field is proposed for the first time. The identified and verified "right flank" asymmetry (5–10%, corresponding to the dominance of right-footed players) serves as an independent criterion for method validity.

Methodological Rigor (25%): 24/25 (assessment for P-Novelty).

- a) **Consistency** — the 2D-AMIS algorithm is strictly described: discretization (40×25), consideration of direction inversion, construction of the transformation function, visualization.
- b) **Scale and representativeness** — the sample of 16 matches (32 team performances) is stratified by tournament standings, ensuring coverage of all game scenarios. The choice of benchmark (2012/13 Russian Championship) is justified from metrological and historical perspectives.
- c) **Transparency** — code is open, data is placed in Dataverse, formulas and step-by-step descriptions are provided.

Point deduction: A formal comparison of AMIS with alternative nonlinear normalization methods (e.g., quantile transformation, Box-Cox) is missing to demonstrate the advantages of the multi-interval approach specifically.

Practical Value (20%): 18/20.

- The method allows direct comparison of spatial game structures of any matches (historical, contemporary) on a unified scale.
- Integration into analysis systems (StatsBomb, Opta) is possible through open libraries.
- Interdisciplinary potential (medicine, urban studies) is argued but not demonstrated with examples.

Visualization Quality (15%): 15/15.

- Figures 1–7 are informative; color scales are unified and correspond to the text.
- Both raw and normalized maps are presented to demonstrate transformation.
- The 40×25 grid with numerical values (Fig. 7) ensures full reproducibility.

Ethical Aspects (10%): 10/10.

- Data deposited in an open repository (Harvard Dataverse).

- Code open (GitHub).
- No conflict of interest identified.
- State support indicated (topic No. 001-22/2).

Supplementary Materials: Present (Dataverse repository, GitHub, Excel data files). In accordance with the prompt requirements, this increases the validity of assessments for novelty and methodology.

Final Score: 28 + 24 + 18 + 15 + 10 = **95/100**

Category: A (fundamental work, new methodology)

Recommendation: Accept (highest standard, 90–100)

Journal Level: Q1

Comments for revision: For publication, it is desirable to add a comparison of AMIS with other nonlinear normalization methods (e.g., quantile transformation) on the same data to formally demonstrate advantages.

Final Comparative Table

No.	Category	Title	Novelty (30)	Methodology (25)	Pract. Value (20)	Visual. (15)	Ethics (10)	Final Score	Recommendation	Journal Level
4	A	AMIS	28	24	18	15	10	95	Accept (highest standard)	Q1
2	Б		20	21	16	12	10	79	Accept with minor revisions	Q2
1	Б		15	18	15	10	10	68	Accept after revision	Q3
3	Г		12	10	8	8	10	48	Reject	—